

2021 AIME II Problems/Problem 9

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Problem

Find the number of ordered pairs (m, n) such that m and n are positive integers in the set $\{1, 2, \dots, 30\}$ and the greatest common divisor of $2^m + 1$ and $2^n - 1$ is not 1.

Solution 1

This solution refers to the **Remarks** section.

By the Euclidean Algorithm, we have

$$\gcd(2^m + 1, 2^m - 1) = \gcd(2, 2^m - 1) = 1.$$

We are given that $\gcd(2^m + 1, 2^n - 1) > 1$. Multiplying both sides by $\gcd(2^m - 1, 2^n - 1)$ gives

$$\begin{aligned} \gcd(2^m + 1, 2^n - 1) \cdot \gcd(2^m - 1, 2^n - 1) &> \gcd(2^m - 1, 2^n - 1) \\ \gcd((2^m + 1)(2^m - 1), 2^n - 1) &> \gcd(2^m - 1, 2^n - 1) && \text{by Claim 1} \\ \gcd(2^{2m} - 1, 2^n - 1) &> \gcd(2^m - 1, 2^n - 1) \\ 2^{\gcd(2m, n)} - 1 &> 2^{\gcd(m, n)} - 1 && \text{by Claim 2} \\ \gcd(2m, n) &> \gcd(m, n), \end{aligned}$$

which implies that n must have more factors of 2 than m does.

We construct the following table for the first 30 positive integers:

# of Factors of 2	Numbers	Count
0	1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29	15
1	2, 6, 10, 14, 18, 22, 26, 30	8
2	4, 12, 20, 28	4
3	8, 24	2
4	16	1

To count the ordered pairs (m, n) , we perform casework on the number of factors of 2 that m has:

1. If m has 0 factors of 2 , then m has 15 options and n has $8 + 4 + 2 + 1 = 15$ options. So, this case has $15 \cdot 15 = 225$ ordered pairs.
2. If m has 1 factor of 2 , then m has 8 options and n has $4 + 2 + 1 = 7$ options. So, this case has $8 \cdot 7 = 56$ ordered pairs.
3. If m has 2 factors of 2 , then m has 4 options and n has $2 + 1 = 3$ options. So, this case has $4 \cdot 3 = 12$ ordered pairs.
4. If m has 3 factors of 2 , then m has 2 options and n has 1 option. So, this case has $2 \cdot 1 = 2$ ordered pairs.

Together, the answer is $225 + 56 + 12 + 2 = \boxed{295}$.

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Solution 2

Consider any ordered pair (m, n) such that $\gcd(2^m + 1, 2^n - 1) > 1$. There must exist some odd number $p \neq 1$ such that $2^m \equiv -1 \pmod{p}$ and $2^n \equiv 1 \pmod{p}$. Let d be the order of 2 modulo p . Note that $2^{2m} \equiv 1 \pmod{p}$. From this, we can say that $2m$ and n are both multiples of d , but m is not. Thus, we have $v_2(n) \geq v_2(d)$ and $v_2(m) + 1 = v_2(d)$. Substituting the latter equation into the inequality before gives $v_2(n) \geq v_2(m) + 1$. Since $v_2(n)$ and $v_2(m)$ are integers, this implies $v_2(n) > v_2(m)$. The rest of the solution now proceeds as in Solution 1.

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Remarks

Claim 1 (GCD Property)

If r , s , and t are positive integers such that $\gcd(r, s) = 1$, then $\gcd(r, t) \cdot \gcd(s, t) = \gcd(rs, t)$.

As r and s are relatively prime (have no prime divisors in common), this property is intuitive.

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Claim 2 (Olympiad Number Theory Lemma)

If u , a , and b are positive integers such that $u \geq 2$, then $\gcd(u^a - 1, u^b - 1) = u^{\gcd(a, b)} - 1$.

There are two proofs to this claim, as shown below.

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Claim 2 Proof 1 (Euclidean Algorithm)

If $a = b$, then $\gcd(a, b) = a = b$, from which the claim is clearly true.

Otherwise, let $a > b$ without the loss of generality. For all integers p and q such that $p > q > 0$, the Euclidean Algorithm states that

$$\gcd(p, q) = \gcd(p - q, q) = \cdots = \gcd(p \bmod q, q).$$

We apply this result repeatedly to reduce the larger number:

$$\gcd(u^a - 1, u^b - 1) = \gcd(u^a - 1 - u^{a-b}(u^b - 1), u^b - 1) = \gcd(u^{a-b} - 1, u^b - 1).$$

Continuing, we have

$$\begin{aligned}\gcd(u^a - 1, u^b - 1) &= \gcd(u^{a-b} - 1, u^b - 1) \\ &\vdots \\ &= \gcd(u^{\gcd(a,b)} - 1, u^{\gcd(a,b)} - 1) \\ &= u^{\gcd(a,b)} - 1,\end{aligned}$$

from which the proof is complete.

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Claim 2 Proof 2 (Bézout's Identity)

Let $d = \gcd(u^a - 1, u^b - 1)$. It follows that $u^a \equiv 1 \pmod{d}$ and $u^b \equiv 1 \pmod{d}$.

By Bézout's Identity, there exist integers x and y such that $ax + by = \gcd(a, b)$, so

$$u^{\gcd(a,b)} = u^{ax+by} = (u^a)^x \cdot (u^b)^y \equiv 1 \pmod{d},$$

from which $u^{\gcd(a,b)} - 1 \equiv 0 \pmod{d}$. We know that $u^{\gcd(a,b)} - 1 \geq d$.

Next, we notice that

$$\begin{aligned}u^a - 1 &= (u^{\gcd(a,b)} - 1)(u^{a-\gcd(a,b)} + u^{a-2\gcd(a,b)} + u^{a-3\gcd(a,b)} + \dots + 1), \\ u^b - 1 &= (u^{\gcd(a,b)} - 1)(u^{b-\gcd(a,b)} + u^{b-2\gcd(a,b)} + u^{b-3\gcd(a,b)} + \dots + 1).\end{aligned}$$

Since $u^{\gcd(a,b)} - 1$ is a common divisor of $u^a - 1$ and $u^b - 1$, we conclude that $u^{\gcd(a,b)} - 1 = d$, from which the proof is complete.

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Video Solution

<https://youtu.be/h3awf7yhGZ4>

~MathProblemSolvingSkills.com

Video Solution by Interstigation

<https://youtu.be/kasgsb0Rge4>

~Interstigation

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